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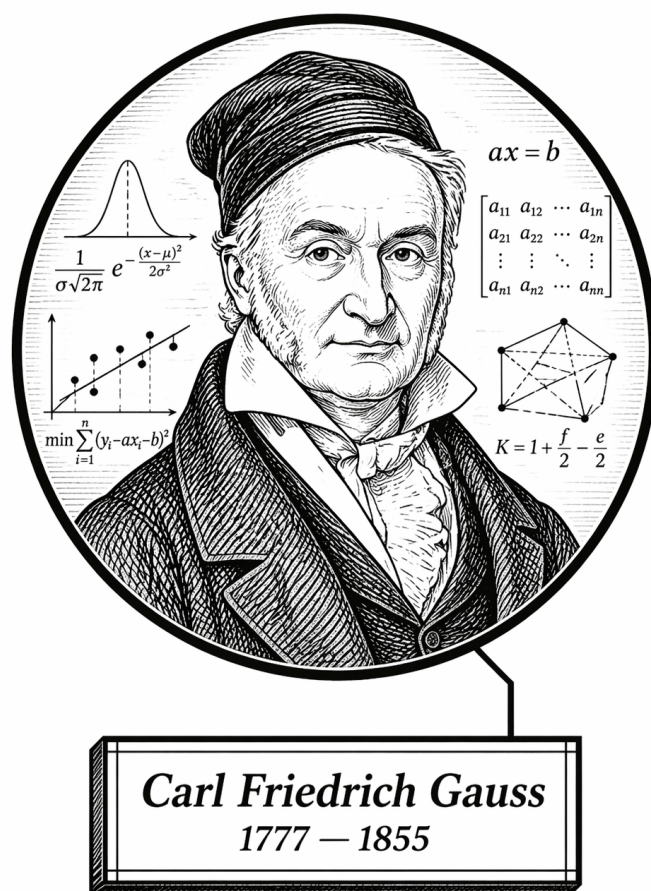
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**Part I**

**Topology and Geometry**

# Volume V

## Topology and Geometry



*"So many squares..."*

– Carl Gauss (probably)

# **Volume V**

Topology and Geometry

*Self-Study Notes*

William Sollers

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# Chapter 1

## Euclidean Geometry

geometry

Geometry  $\rightarrow$  **Euclidean Geometry**  $\rightarrow$  Topology and Geometry

### 1.1 Foundations

#### Definitions

*Remark* (Source note). The following definitions are Euclid's Book I definitions in the Heath translation [Euclid, 1956, Book I, Definitions 1–23].

##### Definition

**Definition 1.1.1** (Point). A point is that which has no part.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

##### Definition

**Definition 1.1.2** (Line). A line is breadthless length.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

##### Definition

**Definition 1.1.3** (Ends of a line). The ends of a line are points.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

##### Definition

**Definition 1.1.4** (Straight line). A straight line is a line which lies evenly with the points on itself.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.5** (Surface). A surface is that which has length and breadth only.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.6** (Edges of a surface). The edges of a surface are lines.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.7** (Plane surface). A plane surface is a surface which lies evenly with the straight lines on itself.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.8** (Plane angle). A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.9** (Rectilinear angle). And when the lines containing the angle are straight, the angle is called rectilinear.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.10** (Right angle and perpendicular). When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

**Definition**

**Definition 1.1.11** (Obtuse angle). An obtuse angle is an angle greater than a right angle.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

**Definition**

**Definition 1.1.12** (Acute angle). An acute angle is an angle less than a right angle.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

**Definition**

**Definition 1.1.13** (Boundary). A boundary is that which is an extremity of anything.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

**Definition**

**Definition 1.1.14** (Figure). A figure is that which is contained by any boundary or boundaries.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

**Definition**

**Definition 1.1.15** (Circle). A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

**Definition**

**Definition 1.1.16** (Center of a circle). And the point is called the center of the circle.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.17** (Diameter). A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.18** (Semicircle). A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.19** (Rectilinear figures). Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.20** (Equilateral, isosceles, and scalene triangles). Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.21** (Right-angled, obtuse-angled, and acute-angled triangles). Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.



*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.22** (Quadrilateral classes). Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Definition

**Definition 1.1.23** (Parallel straight lines). Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

## Postulates

*Remark* (Source note). The following are Euclid's five postulates from Book I of the *Elements* in the Heath translation [Euclid, 1956, Book I, Postulates 1–5].

*Remark* (Exposition). Postulates are geometry-specific starting assumptions. They say which constructions are permitted and which basic geometric facts may be used without proof. In Euclid's system, these are not conclusions reached from earlier results; they are part of the foundation from which the theory begins.

Later propositions are proved from the definitions, the postulates, the common notions, and propositions already established. Thus the postulates are the rules of play for Euclidean geometry: they determine what can be constructed, what may be assumed about straight lines and circles, and where the theory's geometric content enters before proof begins.

#### Axiom

**Axiom 1.1.24** (Euclid Postulate I: Drawing a straight line). Let it have been postulated to draw a straight line from any point to any point.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Axiom

**Axiom 1.1.25** (Euclid Postulate II: Producing a finite straight line). Let it have been postulated to produce a finite straight line continuously in a straight line.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Axiom

**Axiom 1.1.26** (Euclid Postulate III: Describing a circle). Let it have been postulated to describe a circle with any center and distance.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Axiom

**Axiom 1.1.27** (Euclid Postulate IV: Equality of right angles). Let it have been postulated that all right angles are equal to one another.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Axiom

**Axiom 1.1.28** (Euclid Postulate V: Parallel postulate). Let it have been postulated that, if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

## Axioms

*Remark* (Source note). Euclid calls the following principles *common notions*. They function as general axioms governing equality and comparison in Book I of the *Elements* [Euclid, 1956, Book I, Common Notions 1–5].

#### Axiom

**Axiom 1.1.29** (Euclid Common Notion I: Equality is transitive through a common equal). Things which are equal to the same thing are also equal to one another.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

### Axiom

**Axiom 1.1.30** (Euclid Common Notion II: Adding equals to equals). If equals be added to equals, the wholes are equal.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

### Axiom

**Axiom 1.1.31** (Euclid Common Notion III: Subtracting equals from equals). If equals be subtracted from equals, the remainders are equal.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

### Axiom

**Axiom 1.1.32** (Euclid Common Notion IV: Coincidence implies equality). Things which coincide with one another are equal to one another.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

### Axiom

**Axiom 1.1.33** (Euclid Common Notion V: Whole and part). The whole is greater than the part.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

*Remark* (Exposition). Common notions are general principles of reasoning rather than specifically geometric assumptions. They express basic behavior of equality, comparison, and logical transfer: equal things may be substituted for equal things, adding or subtracting equals preserves equality, and a whole exceeds its part.

Because these principles are not tied to points, lines, circles, or planes, they can be used throughout mathematics, not only in geometry. In modern language, they play the role of background axioms governing how equality and comparison may be used inside proofs.

## 1.2 Compass-and-Straightedge Constructions

### Propositions

*Remark* (Source note). The following construction propositions are Euclid's Propositions I.1–I.3 in modern theorem form, following the Heath translation of the *Elements* [Euclid, 1956, Book I, Propositions 1–3].

#### Theorem

**Theorem 1.2.1** (Euclid I.1: Constructing an equilateral triangle). *Given a finite straight line segment  $AB$ , there exists an equilateral triangle  $ABC$  constructed on  $AB$ .*

*Remark* (Proof). Go to proof.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Theorem

**Theorem 1.2.2** (Euclid I.2: Copying a segment from a given point). *Given a point  $A$  and a finite straight line segment  $BC$ , there exists a straight line segment beginning at  $A$  and equal to  $BC$ .*

*Remark* (Proof). Go to proof.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

#### Theorem

**Theorem 1.2.3** (Euclid I.3: Cutting off an equal segment). *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

*Remark* (Proof). Go to proof.

*Remark* (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

# Chapter 2

## Analytical Geometry

geometry

Geometry  $\rightarrow$  **Analytical Geometry**  $\rightarrow$  Topology and Geometry

### 2.1 Analytical Geometry

*Remark* (Status). Notes, proofs, and exercises will appear here in a future revision.

#### Definition

**Definition 2.1.1** (Point-slope form). Let  $P_1 = (x_1, y_1)$  be a point in the Euclidean plane and let  $m \in \mathbb{R}$ . The point-slope form of the line through  $P_1$  with slope  $m$  is

$$y - y_1 = m(x - x_1).$$

*Remark* (Standard quantified statement). For every point  $P_1 = (x_1, y_1) \in \mathbb{R}^2$  and every  $m \in \mathbb{R}$ , the corresponding line is the set of all  $(x, y) \in \mathbb{R}^2$  satisfying  $y - y_1 = m(x - x_1)$ .

*Remark* (Derivation). The formula records the condition that the slope from  $P_1$  to an arbitrary point  $(x, y)$  on the line is constant and equal to  $m$ .

#### Definition

*Remark* (Dependencies).

- Real Numbers
- Cartesian Product
- Subtraction on  $\mathbb{R}$
- Multiplication on  $\mathbb{R}$

**Definition 2.1.2** (Two-point form). Let  $P_1 = (x_1, y_1)$  and  $P_2 = (x_2, y_2)$  be distinct points with  $x_1 \neq x_2$ . The two-point form of the line through  $P_1$  and  $P_2$  is

$$y - y_1 = \left( \frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1).$$

*Remark* (Standard quantified statement). For all distinct points  $P_1 = (x_1, y_1)$  and  $P_2 = (x_2, y_2)$  in  $\mathbb{R}^2$  with  $x_1 \neq x_2$ , the line through  $P_1$  and  $P_2$  is the set of all  $(x, y) \in \mathbb{R}^2$  satisfying the displayed two-point equation.

#### Definition

*Remark* (Dependencies).

- Point-Slope Form
- Subtraction on  $\mathbb{R}$
- $\mathbb{R}$  Is a Field

**Definition 2.1.3** (General linear form). The general linear form of a line in  $\mathbb{R}^2$  is

$$Ax + By + C = 0,$$

where  $A, B, C \in \mathbb{R}$  and  $A^2 + B^2 \neq 0$ .

*Remark* (Standard quantified statement). For all  $A, B, C \in \mathbb{R}$  with  $A^2 + B^2 \neq 0$ , the equation  $Ax + By + C = 0$  determines a nondegenerate line in  $\mathbb{R}^2$ .

*Remark* (Interpretation). The vector  $(A, B)$  is normal to the line. This form includes vertical lines and therefore avoids the exceptional cases present in slope-based forms.

#### Definition

*Remark* (Dependencies).

- Real Numbers
- Addition on  $\mathbb{R}$
- Multiplication on  $\mathbb{R}$
- $\mathbb{R}$  Is a Field

**Definition 2.1.4** (Vector parametric form). A line in  $\mathbb{R}^2$  may be represented as the image of  $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$  given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where  $\mathbf{r}_0 \in \mathbb{R}^2$  is a point on the line and  $\mathbf{v} \in \mathbb{R}^2$  is a nonzero direction vector.

*Remark* (Standard quantified statement). For every  $\mathbf{r}_0 \in \mathbb{R}^2$  and every nonzero  $\mathbf{v} \in \mathbb{R}^2$ , the set  $\{\mathbf{r}_0 + t\mathbf{v} : t \in \mathbb{R}\}$  is a line in  $\mathbb{R}^2$ .

### Definition

*Remark* (Dependencies).

- Real Numbers
- Cartesian Product
- Function
- Addition on  $\mathbb{R}$
- Multiplication on  $\mathbb{R}$

**Definition 2.1.5** (Intercept form). If a line meets the  $x$ -axis at  $(a, 0)$  and the  $y$ -axis at  $(0, b)$  with  $a, b \neq 0$ , its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

*Remark* (Standard quantified statement). For all nonzero  $a, b \in \mathbb{R}$ , the line with intercepts  $(a, 0)$  and  $(0, b)$  is the set of all  $(x, y) \in \mathbb{R}^2$  satisfying  $x/a + y/b = 1$ .

*Remark* (Derivation). The intercept form follows by applying the two-point form to the points  $(a, 0)$  and  $(0, b)$  and then rearranging the resulting equation.

### Definition

*Remark* (Dependencies).

- Two-Point Form
- $\mathbb{R}$  Is a Field

**Definition 2.1.6** (Affine parametric line in  $\mathbb{R}^n$ ). Given distinct points  $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$ , the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \quad t \in \mathbb{R}.$$

*Remark* (Standard quantified statement). For all distinct  $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$ , the set  $\{\mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1) : t \in \mathbb{R}\}$  is the affine line through  $\mathbf{x}_1$  and  $\mathbf{x}_2$ .

*Remark* (Interpretation). For  $0 \leq t \leq 1$ , the same formula describes the line segment joining  $\mathbf{x}_1$  and  $\mathbf{x}_2$ . For arbitrary real  $t$ , it describes the entire affine line.

### Definition

*Remark* (Dependencies).

- Real Numbers
- Cartesian Product
- Subtraction on  $\mathbb{R}$
- Addition on  $\mathbb{R}$
- Multiplication on  $\mathbb{R}$

**Definition 2.1.7** (Normal form). A line in  $\mathbb{R}^2$  may be written in normal form as

$$x \cos \alpha + y \sin \alpha - p = 0,$$

where  $p \geq 0$  is the distance from the origin to the line.

*Remark* (Standard quantified statement). For every  $p \geq 0$  and every real angle  $\alpha$ , the equation  $x \cos \alpha + y \sin \alpha - p = 0$  describes a line whose unit normal direction is  $(\cos \alpha, \sin \alpha)$ .

### Definition

*Remark* (Dependencies).

- [General Linear Form](#)
- Sine and Cosine
- Order on  $\mathbb{R}$

**Definition 2.1.8** (Symmetric form). Given a point  $(x_1, y_1)$  and direction cosines  $(\cos \theta, \sin \theta)$ , the symmetric form of a line is

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r.$$

*Remark* (Standard quantified statement). For every point  $(x_1, y_1)$  and every direction with nonzero displayed denominators, the symmetric equation describes the same line as the parametric equations  $x = x_1 + r \cos \theta$  and  $y = y_1 + r \sin \theta$ .

*Remark* (Dependencies).

- [Vector Parametric Form](#)
- Sine and Cosine
- $\mathbb{R}$  Is a Field



# Chapter 3

## Trigonometry

geometry

Geometry  $\rightarrow$  **Trigonometry**  $\rightarrow$  Topology and Geometry

# Chapter 4

## Non-Euclidean Geometry

geometry

Geometry  $\rightarrow$  **Non-Euclidean Geometry**  $\rightarrow$  Topology and Geometry

### 4.1 Non-Euclidean Geometry

*Remark* (Status). Notes, proofs, and exercises will appear here in a future revision.

# Chapter 5

## Differential Geometry

geometry

Geometry  $\rightarrow$  **Differential Geometry**  $\rightarrow$  Topology and Geometry

### 5.1 Differential Geometry

*Remark* (Status). Notes, proofs, and exercises will appear here in a future revision.

# Chapter 6

## Riemannian Geometry

geometry

Geometry  $\rightarrow$  **Riemannian Geometry**  $\rightarrow$  Topology and Geometry

### 6.1 Riemannian Geometry

*Remark* (Status). Notes, proofs, and exercises will appear here in a future revision.

# Chapter 7

## Topology

topology

Topology and Geometry  $\rightarrow$  **Topology**  $\rightarrow$  Topology and Geometry

### 7.1 Topology

*Remark* (Status). This chapter is planned. Active mathematical content has not yet been added.

# Bibliography

Euclid. *The Thirteen Books of Euclid's Elements*. Dover Publications, New York, second edition, 1956. Translated with introduction and commentary by Thomas L. Heath.